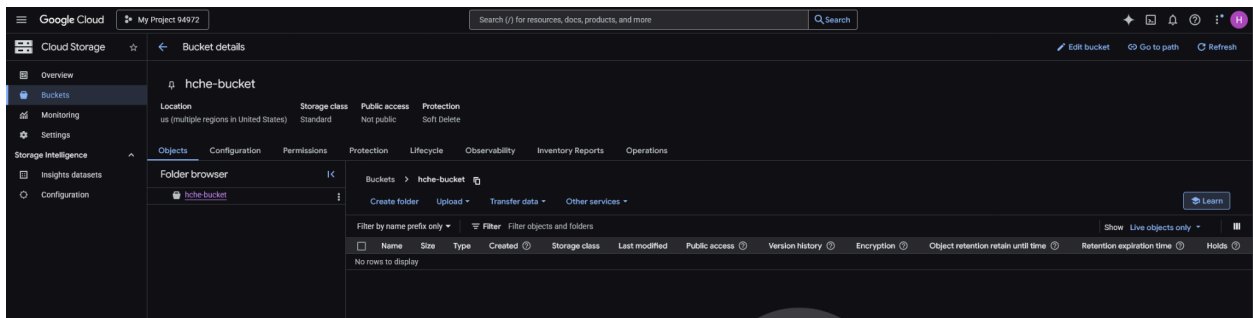
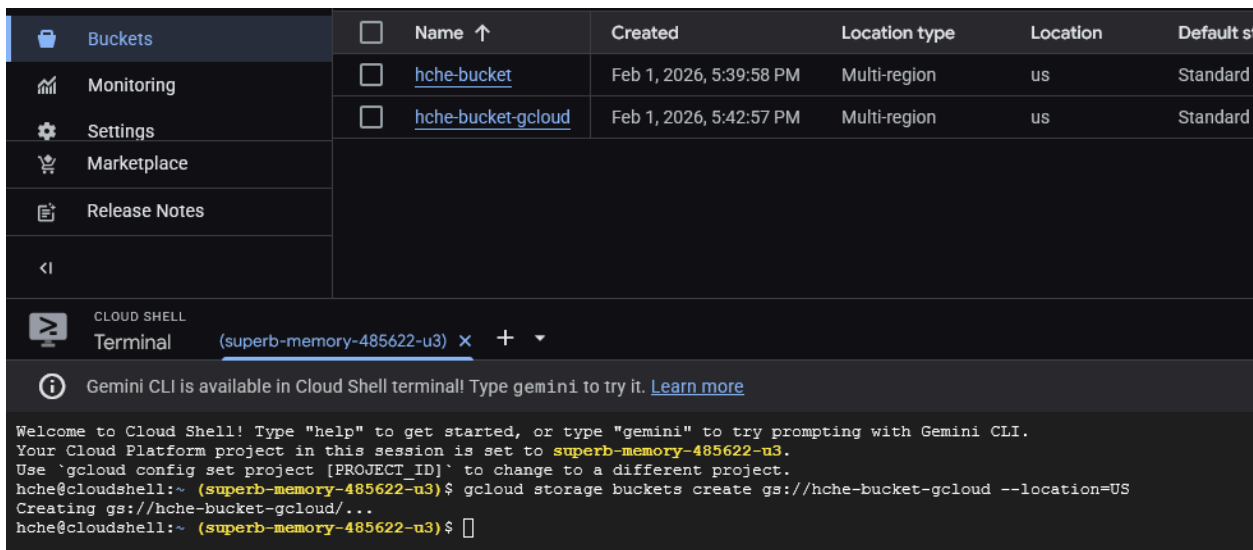


Make a new storage bucket

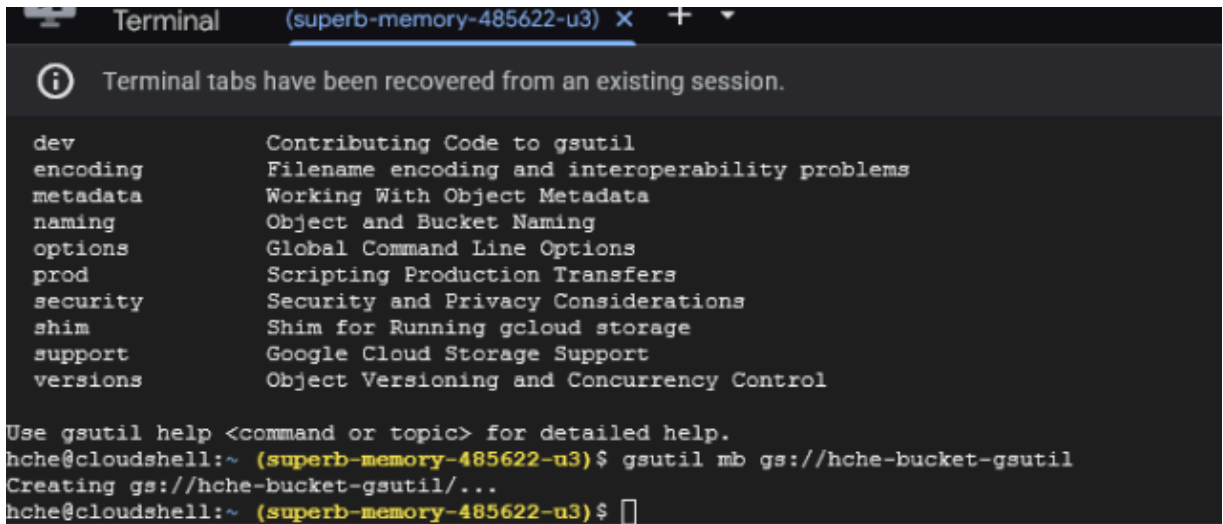
Cloud console



Gcloud



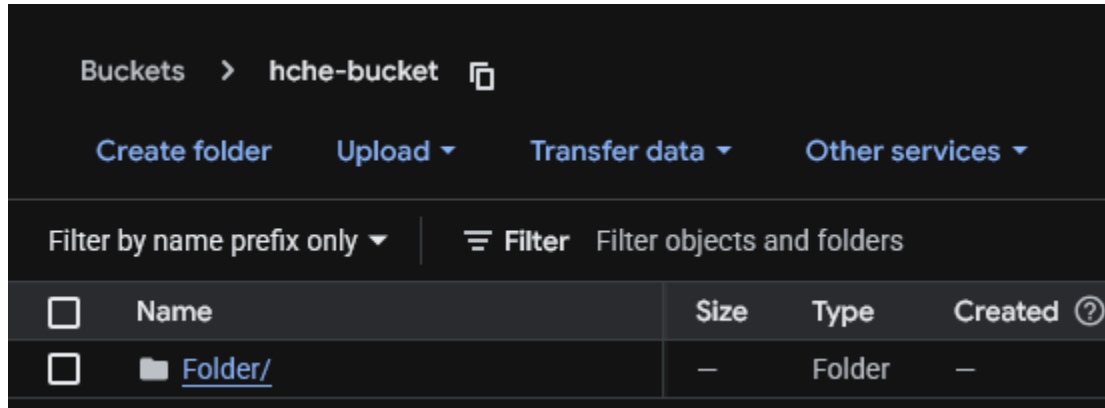
Gsutil



Make a directory inside your bucket

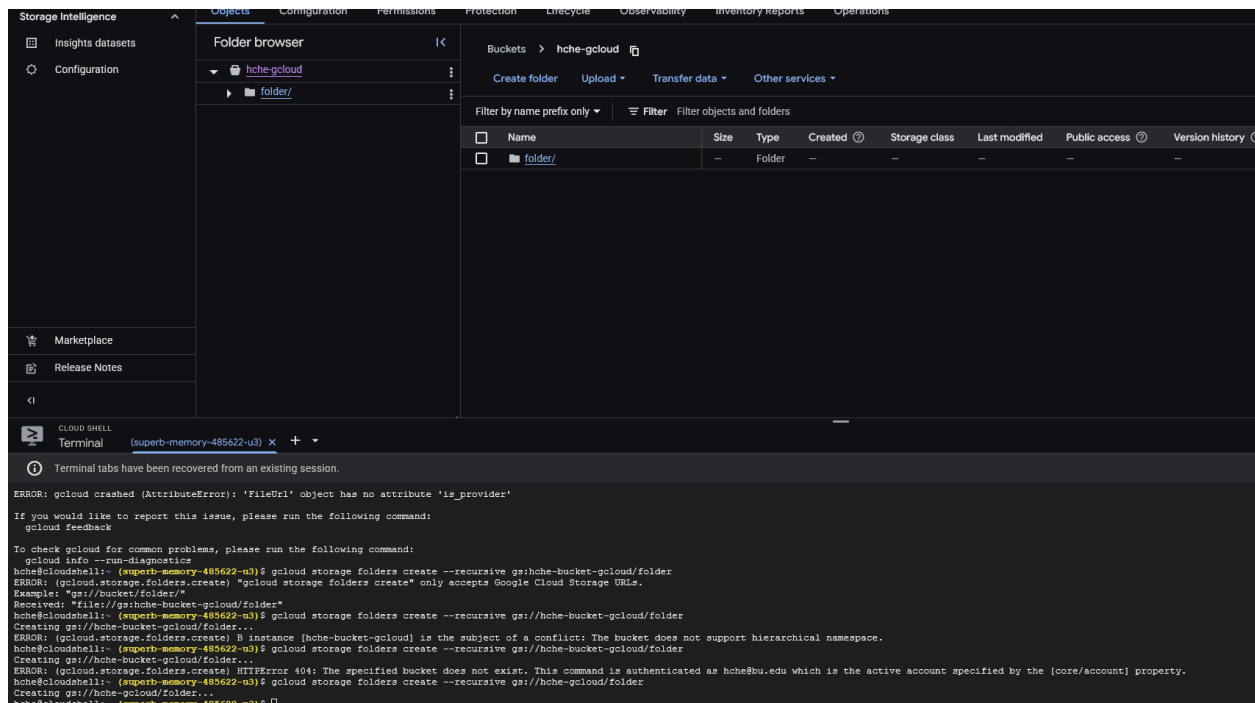
Preface: I deleted and made new buckets to make them use hierarchical name spaces. The console doesn't care if the buckets are flat or hierarchical, but for gcloud and gsutil, you can use roundabout tricks to make folders. You can use a copy command that copies a bucket, and you can add a folder (with an empty file in it, so google knows to create a folder) to the bucket path, effectively making a directory. I don't know if this is what I was supposed to do, but here we are.

Cloud Console



Google Cloud simulates folder structure, but you can create a folder/ directory with the button.

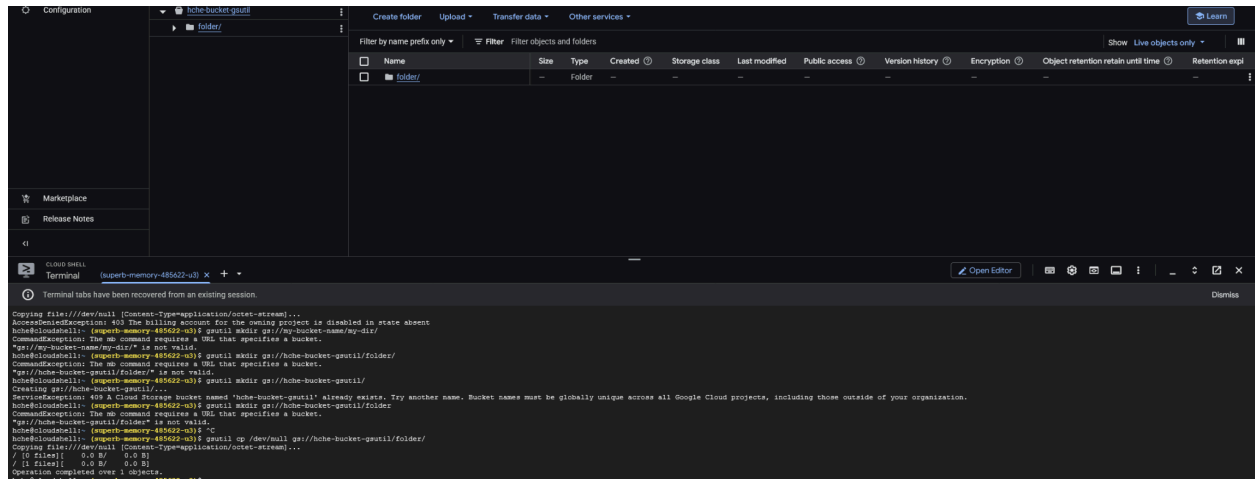
Gcloud



I'm not really sure about Gcloud being able to make a directory but with hierarchical namespace enabled, gcloud storage folders create --recursive gs://BUCKET_NAME/FOLDER_NAME lets you make a folder. But I don't know if this counts. With a flat namespace you can use this gcloud

storage cp /dev/null gs://BUCKET_NAME/FOLDERNAME/ to do the copying strategy that I referred to earlier.

Gsutil



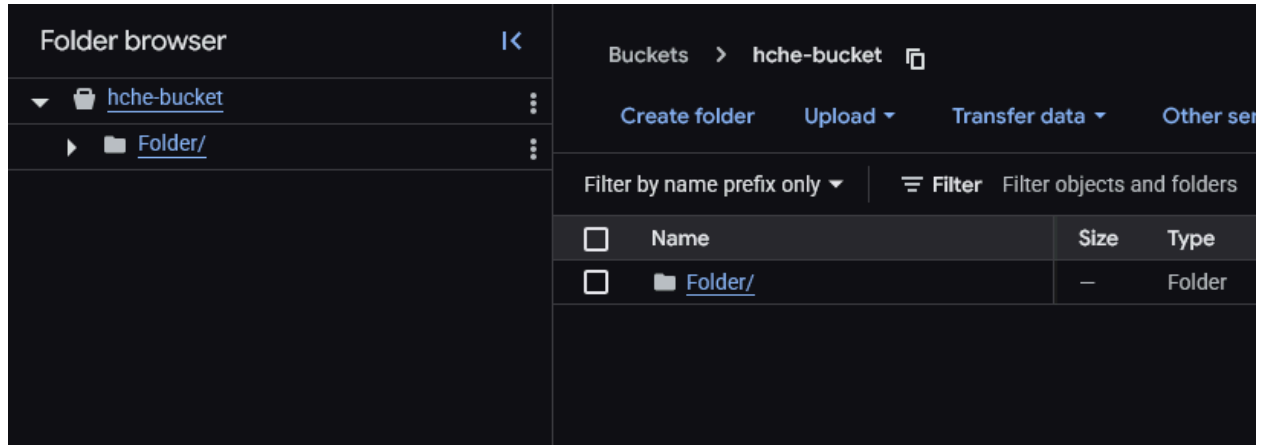
The screenshot shows the Google Cloud Platform console interface. At the top, there's a navigation bar with 'Configuration' and 'hche-bucket-gsutil'. Below this, there's a sidebar with 'Marketplace' and 'Release Notes'. The main area displays a table of objects in the 'hche-bucket-gsutil' bucket. The table has columns for Name, Size, Type, Created, Storage class, Last modified, Public access, Version history, Encryption, Object retention, and Retention expiration. The table shows a single entry for a folder named 'folder/'. Below the table, there's a 'Cloud Shell' terminal window. The terminal shows the following commands and output:

```
Copying file:///dev/null [Content-Type=application/octet-stream]...
hche@cloudshell: (gsutil-memory-485622-us) $ gsutil mkdir gs://hche-bucket-gsutil/folder/
CommandException: The md command requires a URI that specifies a bucket.
"gs://hche-bucket-gsutil/folder/" is not valid.
hche@cloudshell: (gsutil-memory-485622-us) $ gsutil mkdir gs://hche-bucket-gsutil/folder/
CommandException: The md command requires a URI that specifies a bucket.
"gs://hche-bucket-gsutil/folder/" is not valid.
hche@cloudshell: (gsutil-memory-485622-us) $ gsutil mkdir gs://hche-bucket-gsutil/
Creating gs://hche-bucket-gsutil/...
hche@cloudshell: (gsutil-memory-485622-us) $ gsutil mkdir gs://hche-bucket-gsutil/folder/
CommandException: The md command requires a URI that specifies a bucket.
"gs://hche-bucket-gsutil/folder/" is not valid.
hche@cloudshell: (gsutil-memory-485622-us) $
hche@cloudshell: (gsutil-memory-485622-us) $ gsutil cp /dev/null gs://hche-bucket-gsutil/folder/
Copying file:///dev/null [Content-Type=application/octet-stream]...
/ [0 files] 0.0 B/ 0.0 B
/ [1 files] 0.0 B/ 0.0 B
Operation completed over 1 objects.
```

I don't think I can use the Gcloud equivalent command with gsutil even with hierarchical namespace enabled. However, even with it off, you are allowed to simulate it by copying the bucket and creating a zero-byte object with a trailing slash to represent the folder. So by copying the bucket path and uploading more stuff along that path, it'll work as we want it too.

List the directory in your bucket

Cloud Console



Gcloud

```
hche@cloudshell:~ (superb-memory-485622-u3) $ gcloud storage ls gs://hche-gcloud  
gs://hche-gcloud/folder/
```

This one also works.

```
gs://hche-gcloud/folder/" is not valid.  
hche@cloudshell:~ (superb-memory-485622-u3) $ gcloud storage folders list gs://hche-gcloud  
---  
bucket: hche-gcloud  
create_time: 2026-02-01T23:43:55+0000  
metageneration: 1  
name: folder/  
storage_url: gs://hche-gcloud/folder/  
update_time: 2026-02-01T23:43:55+0000
```

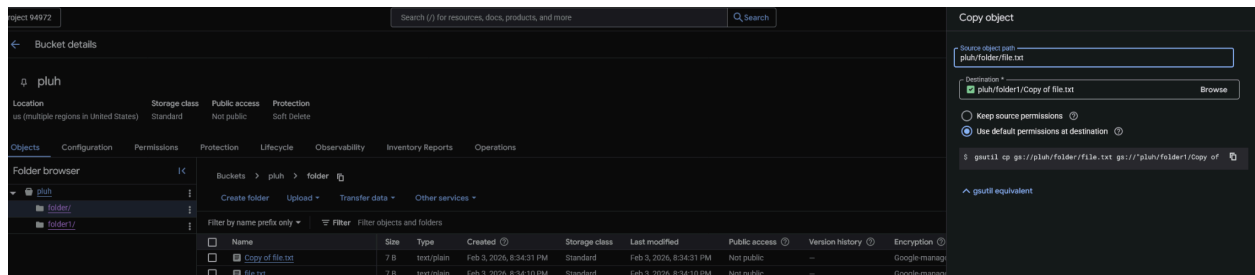
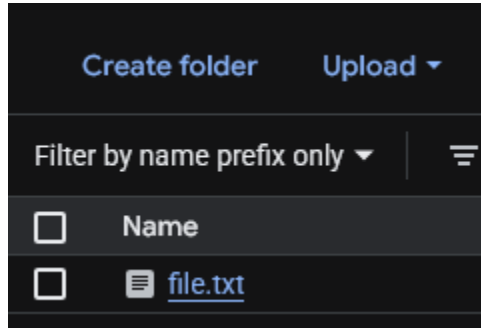
Gsutil

```
hche@cloudshell:~ (superb-memory-485622-u3) $ gsutil ls gs://hche-bucket-gsutil/  
gs://hche-bucket-gsutil/folder/  
hche@cloudshell:~ (superb-memory-485622-u3) $
```

Copy a file into your directory

Cloud Console

First photo is uploading, the second is copying into another directory.



Gcloud

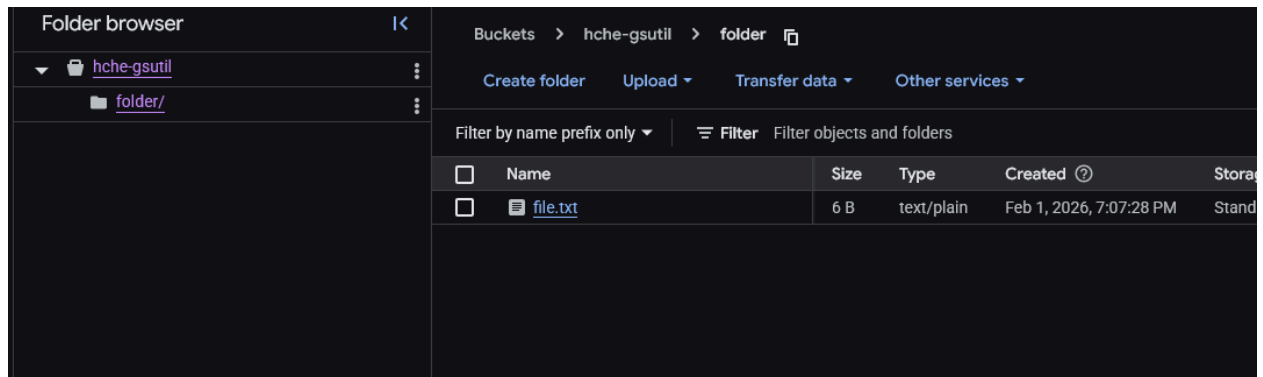
```
hche@cloudshell:~ (superb-memory-485622-u3)$ gcloud storage cp file.txt gs://hche-gcloud/folder
Copying file://file.txt to gs://hche-gcloud/folder
Completed files 1/1 | 6.0B/6.0B
hche@cloudshell:~ (superb-memory-485622-p3)$
```

Gsutil

```
hche@cloudshell:~ (superb-memory-485622-u3)$ gsutil cp file.txt gs://hche-gsutil/folder/
Copying file://file.txt [Content-Type=text/plain]...
/ [1 files][ 6.0 B/ 6.0 B]
Operation completed over 1 objects/6.0 B.
```

List the file inside your directory

Cloud Console



Gcloud

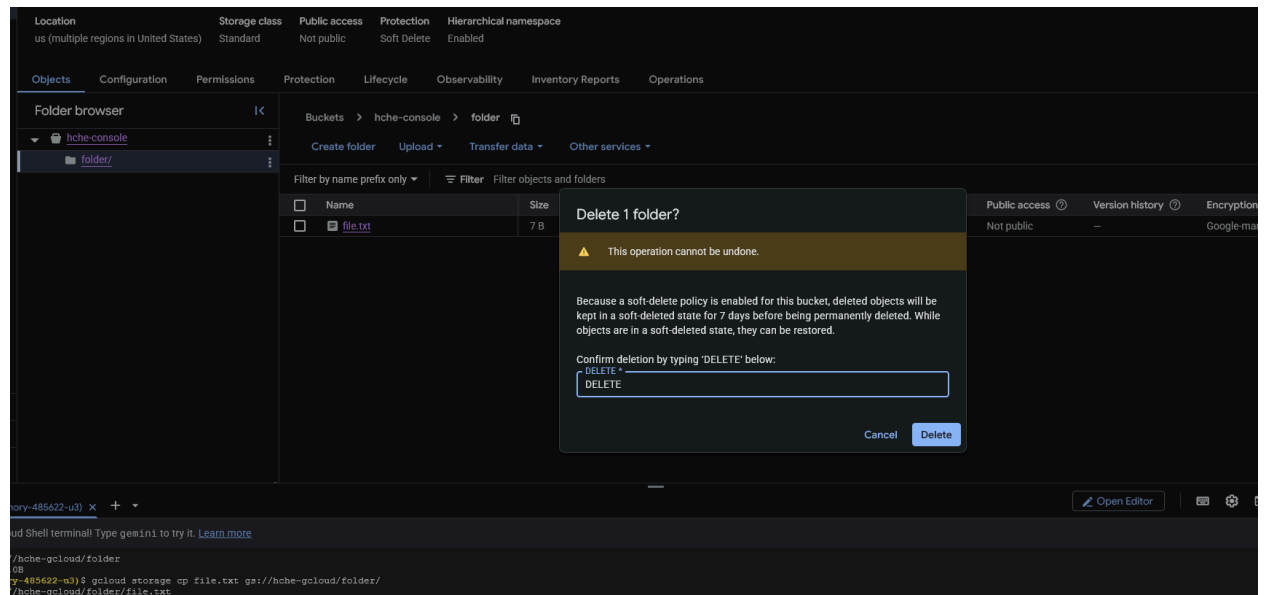
```
hche@cloudshell:~ (superb-memory-485622-u3)$ gcloud storage ls gs://hche-gcloud/folder
gs://hche-gcloud/folder/file.txt
```

Gsutil

```
hche@cloudshell:~ (superb-memory-485622-u3)$ gsutil ls gs://hche-gsutil/folder
gs://hche-gsutil/folder/
gs://hche-gsutil/folder/file.txt
```

Delete the directory (and the file inside it) in a single operation

Cloud Console



Gcloud

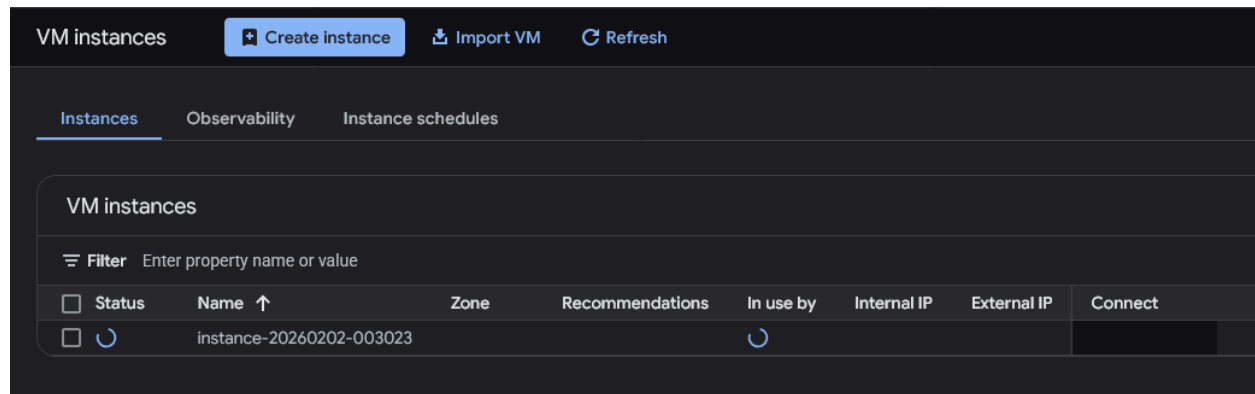
```
Operation completed over 2 objects.
hche@cloudshell:~ (superb-memory-485622-u3) $ gcloud storage rm --recursive gs://hche-gcloud/folder/
Removing objects:
Removing gs://hche-gcloud/folder/file.txt#1769990706656294...
Completed 1/1
Removing folders:
Removing gs://hche-gcloud/folder/...
Completed 1/1
hche@cloudshell:~ (superb-memory-485622-u3) $
```

Gsutil

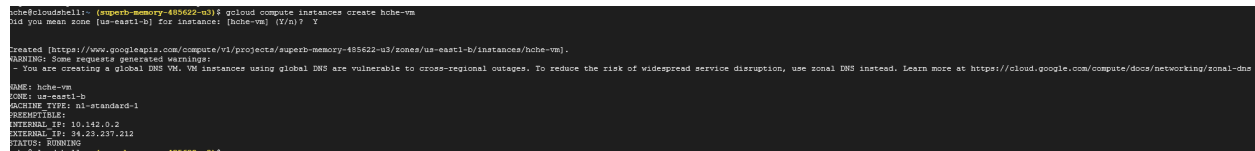
```
gs://hche-gsutil/folder/file.txt
hche@cloudshell:~ (superb-memory-485622-u3) $ gsutil rm -r gs://hche-gsutil/folder/
Removing gs://hche-gsutil/folder/#1769990814660004...
Removing gs://hche-gsutil/folder/file.txt#1769990848143404...
/ [2 objects]
Operation completed over 2 objects.
```

Make a VM

Cloud Console

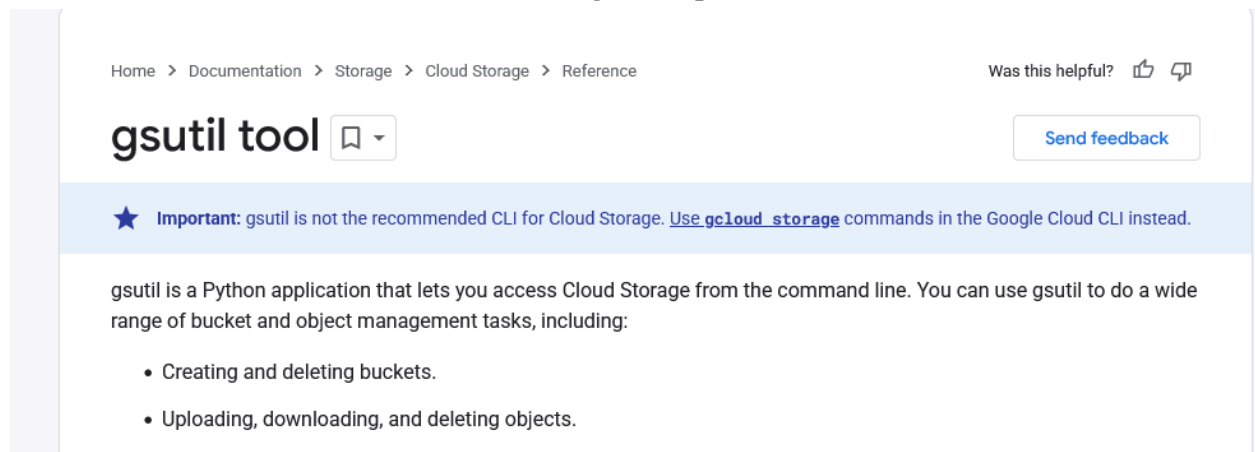


Gcloud



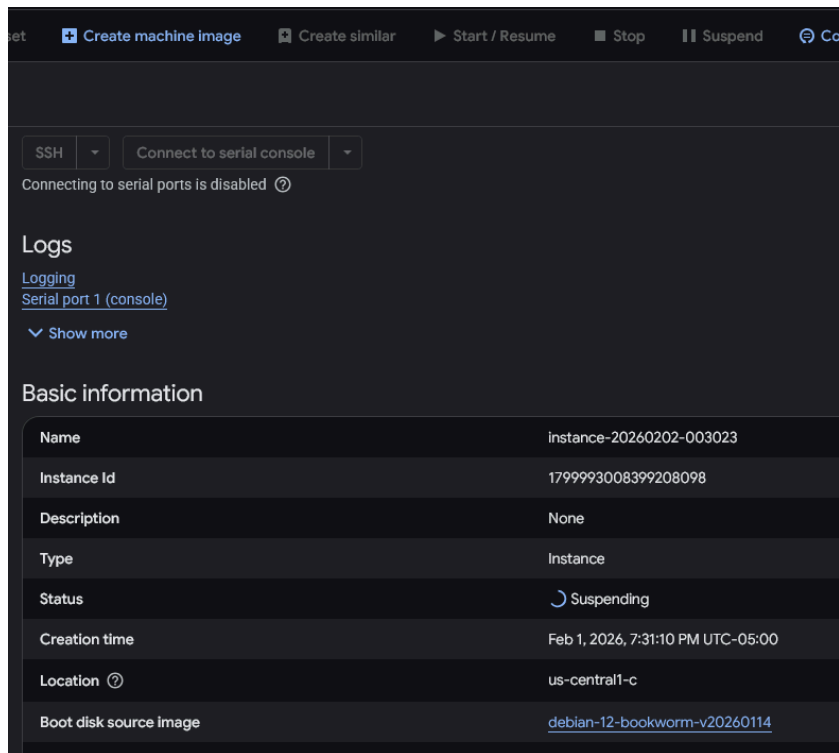
Gsutil

According to the gsutil documentation, it's only meant to be used for Google Cloud storage, thus it can't work with VMs as VMs are under Google Compute.



Suspend the VM

Cloud Console



The screenshot shows the Google Cloud Console interface for a VM instance. At the top, there are buttons for 'Create machine image', 'Create similar', 'Start / Resume', 'Stop', 'Suspend', and 'Co'. Below these, there are tabs for 'SSH' and 'Connect to serial console'. A message states 'Connecting to serial ports is disabled'. The 'Logs' section shows 'Logging' and 'Serial port 1 (console)' with a 'Show more' link. The 'Basic information' section contains a table with the following details:

Name	instance-20260202-003023
Instance Id	1799993008399208098
Description	None
Type	Instance
Status	Suspending
Creation time	Feb 1, 2026, 7:31:10 PM UTC-05:00
Location	us-central1-c
Boot disk source image	debian-12-bookworm-v20260114

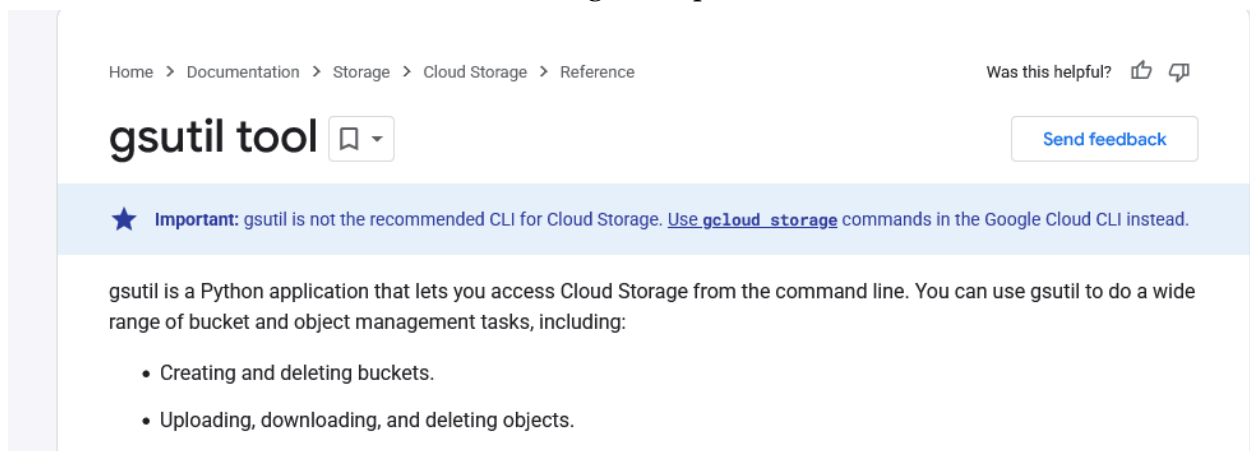
Gcloud

```
hche@cloudshell:~ (superb-memory-485622-u3)$ gcloud compute instances suspend hche-vm --zone=us-central1-a
ERROR: (gcloud.compute.instances.suspend) HTTPError 404: The resource 'projects/superb-memory-485622-u3/zones/us-central1-a/instances/hche-vm' was not found. This resource may not exist.
hche@cloudshell:~ (superb-memory-485622-u3)$ gcloud compute instances suspend hche-vm
Did you mean zone [us-east1-b] for instance: [hche-vm] (Y/n)? Y

Suspending instance(s) hche-vm...done.
Updated [https://compute.googleapis.com/compute/v1/projects/superb-memory-485622-u3/zones/us-east1-b/instances/hche-vm].
```

Gsutil

According to the gsutil documentation, it's only meant to be used for Google Cloud storage, thus it can't work with VMs as VMs are under Google Compute.



The screenshot shows the Google Cloud documentation page for the 'gsutil tool'. The breadcrumb navigation is 'Home > Documentation > Storage > Cloud Storage > Reference'. There is a 'Was this helpful?' section with thumbs up/down icons and a 'Send feedback' button. The title 'gsutil tool' is followed by a bookmark icon. A blue banner with a star icon contains the text: 'Important: gsutil is not the recommended CLI for Cloud Storage. Use gcloud storage commands in the Google Cloud CLI instead.' Below this, the text states: 'gsutil is a Python application that lets you access Cloud Storage from the command line. You can use gsutil to do a wide range of bucket and object management tasks, including:' followed by a list of tasks:

- Creating and deleting buckets.
- Uploading, downloading, and deleting objects.

Stop the VM

Cloud Console

The screenshot shows the Google Cloud Console interface. At the top, there are tabs for 'Instances', 'Observability', and 'Instance schedules'. Below this, the 'VM instances' section is visible. A table lists VM instances with columns for Status, Name, Zone, Recommendations, In use by, Internal IP, External IP, and Connect. One instance, 'instance-20260202-003023', is highlighted. Below the table, there are 'Related actions' such as 'Protect with Backup & DR', 'View billing report', and 'Monitor VMs'. A modal dialog is open, asking 'Stop instance-20260202-003023?'. It lists resources that will be billed (Persistent disks, Static IP addresses) and provides a warning that the VM will shut down. There is a checkbox for 'Skip graceful shutdown (if applicable)' and a 'Beta' label. Buttons for 'Cancel' and 'Stop' are at the bottom right of the dialog.

Gcloud

```
hche@cloudshell:~ (superb-memory-485622-u3) $ gcloud compute instances stop hche-vm --zone=us-central1-a
ERROR: (gcloud.compute.instances.stop) HTTPError 404: The resource 'projects/superb-memory-485622-u3/zones/us-central1-a/instances/hche-vm' does not exist.
hche@cloudshell:~ (superb-memory-485622-u3) $ gcloud compute instances stop hche-vm
Did you mean zone [us-east1-b] for instance: [hche-vm] (Y/n)? Y

Stopping instance(s) hche-vm...done.
Updated [https://compute.googleapis.com/compute/v1/projects/superb-memory-485622-u3/zones/us-east1-b/instances/hche-vm].
hche@cloudshell:~ (superb-memory-485622-u3) $
```

Gsutil

According to the gsutil documentation, it's only meant to be used for Google Cloud storage, thus it can't work with VMs as VMs are under Google Compute.

The screenshot shows the Google Cloud documentation page for the 'gsutil tool'. The breadcrumb navigation is 'Home > Documentation > Storage > Cloud Storage > Reference'. There is a 'Was this helpful?' section with thumbs up/down icons and a 'Send feedback' button. The title 'gsutil tool' is followed by a dropdown menu. A blue banner with a star icon states: 'Important: gsutil is not the recommended CLI for Cloud Storage. Use [gcloud storage](#) commands in the Google Cloud CLI instead.' Below this, the text explains that 'gsutil is a Python application that lets you access Cloud Storage from the command line. You can use gsutil to do a wide range of bucket and object management tasks, including:'. A bulleted list follows: 'Creating and deleting buckets.' and 'Uploading, downloading, and deleting objects.'.

List all the VMs

Cloud Console

VM instances							
Filter Enter property name or value							
☐ Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☐	instance-20260202-003023	us-central1-c			10.128.0.2 (nic0)		SSH ▾ ⋮
☐	instance-20260202-004002						

Gcloud

```
hche@cloudshell:~ (superb-memory-485622-u3)$ gcloud compute instances list
NAME: hche-vm
ZONE: us-east1-b
MACHINE_TYPE: n1-standard-1
PREEMPTIBLE:
INTERNAL_IP: 10.142.0.2
EXTERNAL_IP:
STATUS: TERMINATED
hche@cloudshell:~ (superb-memory-485622-u3)$
```

Gsutil

According to the gsutil documentation, it's only meant to be used for Google Cloud storage, thus it can't work with VMs as VMs are under Google Compute.

Home > Documentation > Storage > Cloud Storage > Reference

Was this helpful?

gsutil tool

▾

Send feedback

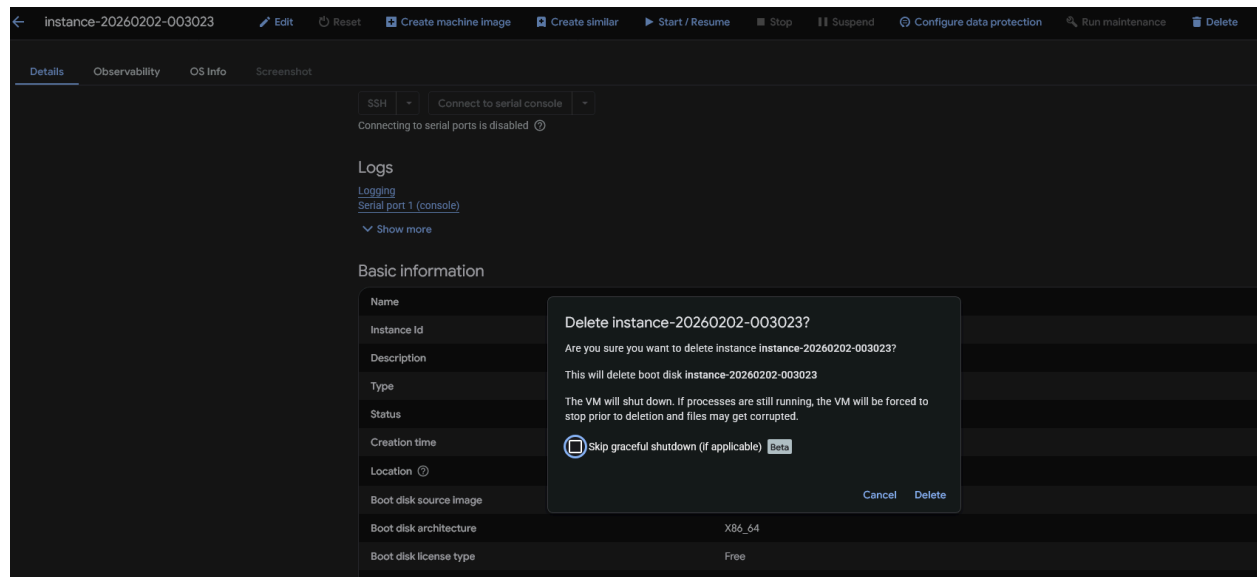
★ **Important:** gsutil is not the recommended CLI for Cloud Storage. Use [gcloud storage](#) commands in the Google Cloud CLI instead.

gsutil is a Python application that lets you access Cloud Storage from the command line. You can use gsutil to do a wide range of bucket and object management tasks, including:

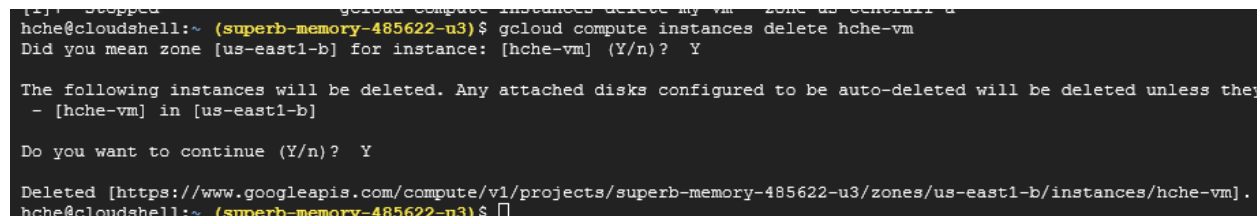
- Creating and deleting buckets.
- Uploading, downloading, and deleting objects.

Delete the VM

Cloud Console

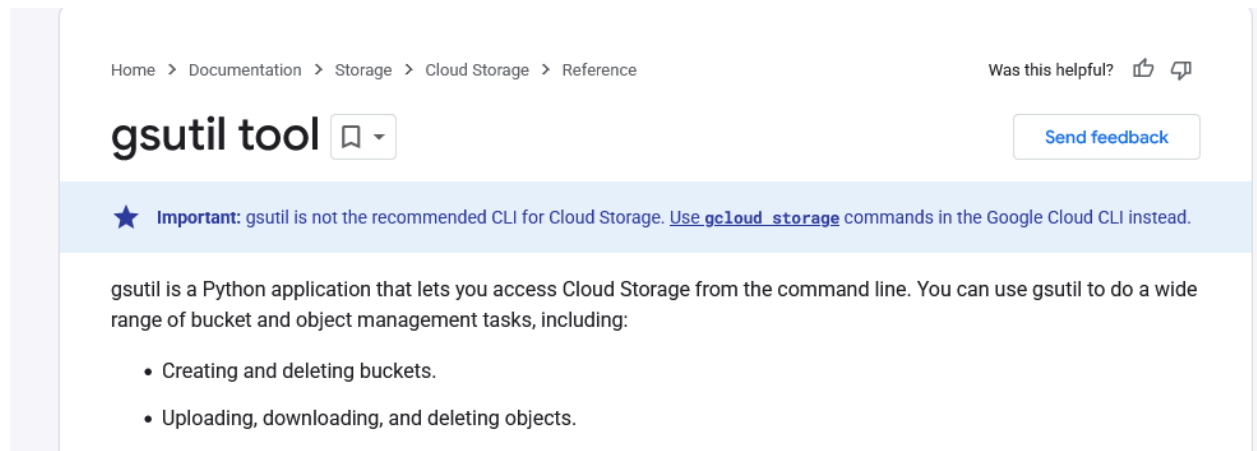


Gcloud



Gsutil

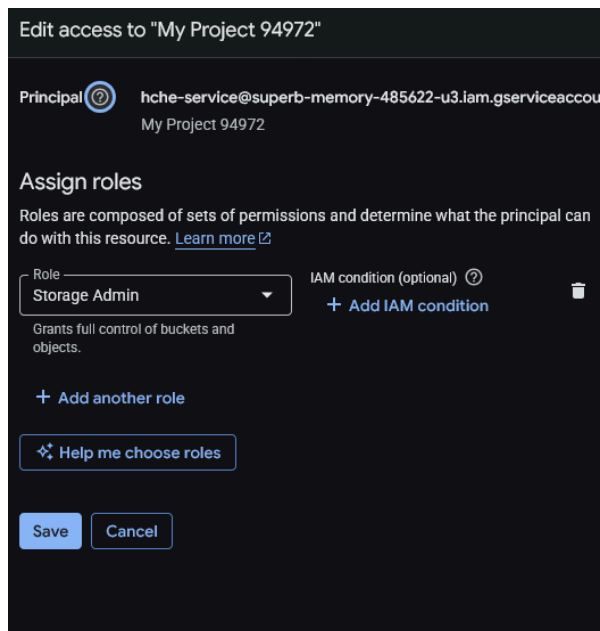
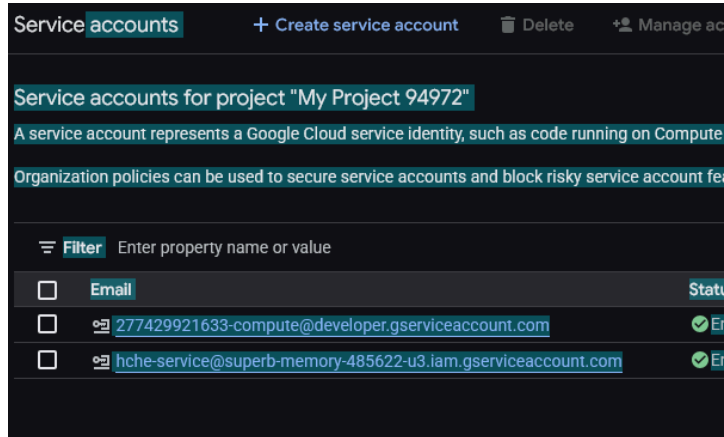
According to the gsutil documentation, it's only meant to be used for Google Cloud storage, thus it can't work with VMs as VMs are under Google Compute.



Make a service account and give it permissions to access your storage bucket

Cloud Console

Can click create service account



Gcloud

```

hche@cloudshell:~ (superb-memory-485622-u3)$ gcloud iam service-accounts create hche-service2
Created service account [hche-service2].
hche@cloudshell:~ (superb-memory-485622-u3)$ gcloud projects add-iam-policy-binding superb-memory-485622-u3 \
--member="serviceAccount:hche-service2@superb-memory-485622-u3.iam.gserviceaccount.com" \
--role="roles/storage.admin"
Updated IAM policy for project [superb-memory-485622-u3].
bindings:
- members:
  - serviceAccount:277429921633@cloudservices.gserviceaccount.com
    role: roles/compute.instanceGroupManagerServiceAgent
- members:
  - serviceAccount:service-277429921633@compute-system.iam.gserviceaccount.com
    role: roles/compute.serviceAgent
- members:
  - serviceAccount:277429921633-compute@developer.gserviceaccount.com
    role: roles/editor
- members:
  - user:hche@bu.edu
    role: roles/owner
- members:
  - serviceAccount:hche-service2@superb-memory-485622-u3.iam.gserviceaccount.com
  - serviceAccount:hche-service@superb-memory-485622-u3.iam.gserviceaccount.com
    role: roles/storage.admin
etag: BwZJzPk10pc=
version: 1
hche@cloudshell:~ (superb-memory-485622-u3)$

```

Gsutil

According to the gsutil documentation, it's only meant to be used for Google Cloud storage, thus it can't work with service accounts as it's under Identity and Access Management (IAM).

Home > Documentation > Storage > Cloud Storage > Reference
Was this helpful?

gsutil tool

Send feedback

★ **Important:** gsutil is not the recommended CLI for Cloud Storage. [Use gcloud storage](#) commands in the Google Cloud CLI instead.

gsutil is a Python application that lets you access Cloud Storage from the command line. You can use gsutil to do a wide range of bucket and object management tasks, including:

- Creating and deleting buckets.
- Uploading, downloading, and deleting objects.

List all your service accounts

Cloud Console

☐	Email	Status	Name ↑	Description	Key ID
☐	 277429921633-compute@developer.gserviceaccount.com	✔ Enabled	Compute Engine default service account		No keys
☐	 hche-service@superb-memory-485622-u3.iam.gserviceaccount.com	✔ Enabled	hche-service		No keys

Gcloud

```
hche@cloudshell:~ (superb-memory-485622-u3)$ gcloud iam service-accounts list
DISPLAY NAME: Compute Engine default service account
EMAIL: 277429921633-compute@developer.gserviceaccount.com
DISABLED: False

DISPLAY NAME:
EMAIL: hche-service2@superb-memory-485622-u3.iam.gserviceaccount.com
DISABLED: False
```

Gsutil

According to the gsutil documentation, it's only meant to be used for Google Cloud storage, thus it can't work with service accounts as it's under Identity and Access Management (IAM).

[Home](#) > [Documentation](#) > [Storage](#) > [Cloud Storage](#) > [Reference](#)Was this helpful?  

gsutil tool

[Send feedback](#)

★ **Important:** gsutil is not the recommended CLI for Cloud Storage. Use [gcloud storage](#) commands in the Google Cloud CLI instead.

gsutil is a Python application that lets you access Cloud Storage from the command line. You can use gsutil to do a wide range of bucket and object management tasks, including:

- Creating and deleting buckets.
- Uploading, downloading, and deleting objects.

Delete the service account

Cloud Console

The screenshot shows the Google Cloud Console interface. At the top, there's a table of service accounts. Below it, a modal dialog titled "Delete service account 'hche-service'" is open. The dialog contains the following text:

Deleting this service account will delete all associated key IDs, and will prevent the account from authenticating to any API.

Deleting this service account will:

- Prevent any Google Cloud service from authenticating with it.
- Remove all key IDs associated with it.
- May cause unexpected interruptions.

You cannot restore or roll back this change easily. We highly recommend disabling the service account first, testing for any unexpected impact, and only then deleting.

At the bottom of the dialog are three buttons: "Cancel", "Disable", and "Delete".

Email	Status	Name	Description	Key ID	Key creation date	OAuth 2 Client ID	Actions
277429921633-compute@developer.gserviceaccount.com	Enabled	Compute Engine default service account		No keys		11155813314679345150	
hche-service@superb-memory-485622-u3.iam.gserviceaccount.com	Enabled	hche-service		No keys		113918612695867539911	

Gcloud

```
hche@cloudshell:~ (superb-memory-485622-u3) $ gcloud iam service-accounts delete hche-service2@superb-memory-485622-u3.iam.gserviceaccount.com
You are about to delete service account [hche-service2@superb-memory-485622-u3.iam.gserviceaccount.com]
Do you want to continue (Y/n)? Y
```

Gsutil

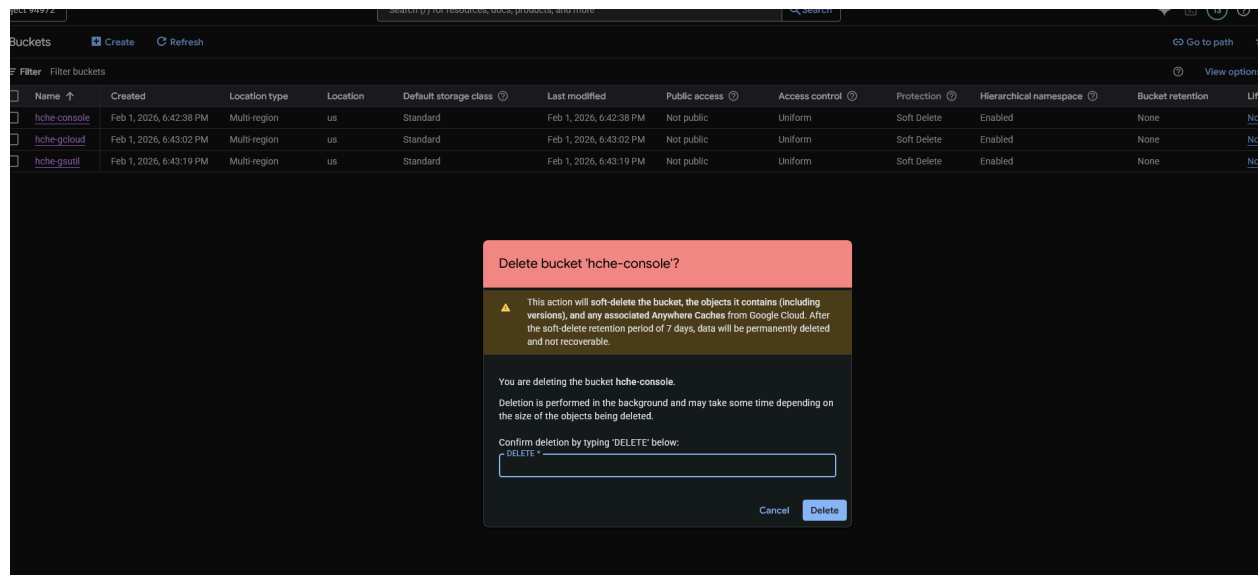
According to the gsutil documentation, it's only meant to be used for Google Cloud storage, thus it can't work with service accounts as it's under Identity and Access Management (IAM).

The screenshot shows the Google Cloud documentation page for the gsutil tool. The breadcrumb navigation is: Home > Documentation > Storage > Cloud Storage > Reference. The page title is "gsutil tool". There is a "Send feedback" button. A blue banner with a star icon contains the text: "Important: gsutil is not the recommended CLI for Cloud Storage. Use gcloud storage commands in the Google Cloud CLI instead." Below this, the text states: "gsutil is a Python application that lets you access Cloud Storage from the command line. You can use gsutil to do a wide range of bucket and object management tasks, including:" followed by a list of tasks:

- Creating and deleting buckets.
- Uploading, downloading, and deleting objects.

Delete the storage bucket

Cloud Console



Gcloud

```
hche@cloudshell:~ (superb-memory-485622-u3)$ gcloud storage buckets delete gs://hche-gcloud
Removing gs://hche-gcloud/...
Completed 1/1
hche@cloudshell:~ (superb-memory-485622-u3)$
```

Gsutil

```
hche@cloudshell:~ (superb-memory-485622-u3)$ gsutil rm -r gs://hche-gsutil
Removing gs://hche-gsutil/...
hche@cloudshell:~ (superb-memory-485622-u3)$
```